

Monty Hall Shock Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	Time	30-40 min
Difficulty	Medium	ID	monty-hall-shock

Big idea: Formal probability can beat intuition.

One-Minute Teacher Brief

Students feel the conflict between intuition and formal probability.

Student Prompt

Inspect Monty Hall Shock Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Monty Hall Shock Stimulus A	Student-facing stimulus 1: Three paper doors, one prize card, and repeated rounds with switch/stay tracking. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Monty Hall Shock Stimulus B	Student-facing stimulus 2: A class results table showing wins after switching versus staying. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Monty Hall Shock Reveal / Twist	Student-facing stimulus 3: A probability tree showing that the initial choice has probability $\frac{1}{3}$ and the two unchosen doors together have probability $\frac{2}{3}$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Formal probability can beat intuition.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Monty Hall Shock Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Monty Hall Shock Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Why does mathematical reasoning often feel wrong at first?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how formal probability can beat intuition.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Why does mathematical reasoning often feel wrong at first?

Setup Checklist

- Prepare a starter stimulus using: Three paper doors, one prize card, and repeated rounds with switch/stay tracking.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- Three paper doors, one prize card, and repeated rounds with switch/stay tracking.
- A class results table showing wins after switching versus staying.
- A probability tree showing that the initial choice has probability $\frac{1}{3}$ and the two unchosen doors together have probability $\frac{2}{3}$.

Resources To Use

- Resource pack: <resources/monty-hall-shock/index.html>
- Card deck CSV: <resources/monty-hall-shock/cards.csv>
- Visual prompt: <resources/monty-hall-shock/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Play the Monty Hall game physically. A student chooses a door.
2. The host opens an empty door.
3. The student chooses whether to stay or switch.
4. Repeat many times and compare win rates for stay and switch.

Debrief Moves

- Knowledge question: Why does mathematical reasoning often feel wrong at first?
- Should we trust intuition or formal proof?
- What makes a mathematical explanation convincing?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...